MIL-T-3595B

14 July 1969
SUPERSEDING
MIL-T-3595A
24 March 1952

MILITARY SPECIFICATION

TUBING, PHOSPHOR BRONZE: (CDA NO. 510) ROUND, SEAMLESS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers round seamless phosphor bronze tubing, furnished in one composition and light and hard drawn tempers.

1.2 Classification -

1.2.1 *Types* - The tubing furnished under this specification shall be of the following types, as specified (see 6.2).

Type 1 - Light Drawn (1/4 Hard)
Type 2 - Hard Drawn

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-3993	Copper and Copper-Base Alloy Mill Products, Packaging of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 4710

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STANDARDS

Federal

FED-STD-146	Tolerances for Copper and Copper Base Alloy Mill Products
Fed. Test Method Std. No. 151	Metals; Test Methods
FED-STD-185	Continuous Identification Marking of Copper and Copper Base Alloy Mill Products

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material - The raw material used shall be such as to produce tubing in accordance with this specification.

3.2 Manufacture - The tubing shall be seamless.

3.3 Chemical composition - The chemical composition of the tubes shall be as shown in Table I. Analysis shall regularly be made only for the elements specifically mentioned in Table I. If the presence of other elements is suspected, or indicated in the course of routine analysis, further analysis shall be made either chemically or spectro-chemically to determine that the total of the other elements is not in excess of the percentage allowed.

TABLE I

CHEMICAL COMPOSITION

TIN	PHOSPHORUS	IRON	LEAD	ZINC	COPPER PLUS TIN PLUS PHOSPHORUS, MINIMUM
PERCENT					
3.5-5.8	0.03-0.35	0.10 1/	0.05 1/	0.30 1/	99.5

1/ Maximum

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3.4 Tensile properties - The tensile properties at room temperature shall be as shown in Table II.

TABLE II

TENSILE PROPERTIES

TYPE	TEMPER	TENSILE STRENGTH (p.s.i)	
		MINIMUM	MAXIMUM
Type 1	Light Drawn (1/4 Hard)	55,000	70,000
Type 2	Hard Drawn	70,000	85,000

3.5 Dimensional tolerances -

3.5.1 Wall thickness - When ordered to specific wall thickness (see 6.2), the tubes shall not vary from the ordered wall thickness by more than the amounts shown in 20a (2) of FED-STD-146.

3.5.2 Average diameter - The average diameter of the tube shall not vary by more than the amounts shown in 20b (1) of FED-STD-146.

3.5.3 When tube is ordered by outside diameter and inside diameter, the maximum deviation of the wall thickness from the nominal wall thickness shall not exceed by more than 50 percent the permissible variations shown in 20a (2) of FED-STD-146.

3.5.4 Length -

3.5.4.1 Specific lengths - When ordered to specific lengths, tubes shall be not less than the ordered specific length and shall not exceed it by more than the amounts shown in 20a (3) of FED-STD-146 (see 6.2).

3.5.4.2 Specific lengths with ends (short lengths) included - When tubes are ordered to specific lengths with ends (short lengths) included, the length of each piece, except the short lengths, shall be not less than ordered length and shall not exceed it by more than 1 inch. The permissible quantity and length of ends (short pieces) in any shipment shall be as shown in 20a (4) of FED-STD-146 (see 6.2).

3.5.4.3 Stock lengths, and stock lengths with ends (short lengths) included - When tubes are ordered in stock lengths or in stock lengths with ends (short lengths) included, the length of each piece, except short lengths, shall be not less than the

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ordered stock length and shall not exceed it by more than 1 inch. The permissible quantity and length of ends (short pieces) in any shipment shall be as shown in 20a (4) of FED-STD-146 (see 6.2).

3.5.5 Squareness of end cuts - The departure from squareness of end cuts of tubes shall not exceed limits shown in 20a (6) of FED-STD-146.

3.5.6 Straightness - The maximum depth of arc of tubes ordered in straight lengths shall not exceed limits shown in 20a (5) of FED-STD-146.

3.5.7 Roundness - For tubes ordered in straight lengths, the difference between major and minor diameters shall not exceed limits shown in 20a (7) of FED-STD-146.

3.6 Cleaning - All tubes shall be thoroughly cleaned by pickling or other suitable method before being presented for inspection.

3.7 Identification marking - Tubing shall be marked in accordance with FED-STD-185. The specification number, alloy number, and temper designation shall be included in the identification marking.

3.8 Workmanship - All tubes shall be sound, of uniform quality and condition, free from excessive nonmetallic inclusions, cracks, slivers, twists, seams, grooves, scale, damaged ends, buckles and other injurious defects. Inside and outside surfaces shall be sufficiently free from irregularities to finish smooth inside and out by machining cuts not to exceed 1/32 inch in depth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot - Unless otherwise specified by the procuring activity, a lot shall consist of all tubes of the same size submitted for inspection at the same time, and shall not exceed 5,000 pounds.

4.3 Samples for chemical analysis - A sample for Government analysis shall be taken from each lot. The inspector shall select samples from 0.5 percent of the number of pieces in the lot, except that not less than two nor more than ten samples shall be taken from any lot. From the pieces selected, equal amounts of

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clean clippings, drillings, or millings, shall be obtained, combined, and thoroughly mixed. The inspector shall forward for analysis not less than 2 ounces of the resulting mixture or an adequate solid sample from each sample selected. The chemical analysis shall be by the wet chemical method in accordance with Method III or by the spectro-chemical method in accordance with Method 112 of Fed. Test Method Std. No. 151. In case of dispute, the wet method shall be used.

4.4 Tension tests -

4.4.1 Samples - Two tension samples shall be selected by the inspector from different pieces in each lot.

4.4.2 Specimens - Tension test specimens shall be the full cross-section of the tube where practicable. In cases where machined specimens are necessary, the specimens shall conform to Type 5, Fed. Test Method Std. No. 151. The axis of the specimen shall be parallel to the direction of rolling.

4.4.3 Method - Tension tests shall be conducted in accordance with Fed. Test Method Std. No. 151.

4.5 Inspection - All tubes shall be surface inspected for workmanship, and a sufficient number of measurements made to determine compliance with the dimensional requirements. Wall thickness, outside diameter, and inside diameter shall be determined by micrometer measurement.

4.6 Inspection of preparation for delivery - An inspection shall be made to determine that packing and marking comply with the requirements in Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing - Packing shall be Level A, B, or C as specified (see 6.2).

5.1.1 Levels A and B - The material shall be packed in accordance with MIL-C-3993.

5.1.2 Level C - The products shall be separated by size, composition and temper and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.2 Marking - In addition to any special marking required (see 6.2 and 6.3) by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use - This material is used for the manufacture of cannon components by machining all surfaces of the tubing. It is not intended for ordinary tubing uses such as the conduction of fluids.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of the specification.

(b) Type and size of tubing required (Two of the following should be specified: inside diameter, outside diameter, wall thickness). (See 1.2.1, 3.5.1, and 3.5.3).

(c) Level of packing required (see 5.1) and maximum gross weight of container.

(d) Identification marking in addition to requirements of FED-STD-185 (see 3.7).

(e) Special marking, if required (see 5.2).

(f) Whether specific lengths, specific lengths with ends, or stock lengths and stock lengths with ends are required (see 3.5.4.1, 3.5.4.2, and 3.5.4.3).

6.3 Applicability of marking and packing requirements - The requirements for identification marking (see 3.7), for packing (see 5.1) and marking for shipment (see 5.2) apply to direct shipment to Government activities and apply also to shipments between the manufacturer and the government prime contractor.

Custodians:
Army - MR
Navy - AS
Air Force - 11

Review activities:
Air Force - 82, 85

User activities:
Army - ME
Navy - OS

Preparing activity:
Navy - AS
Project No. 4710-0133

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